

CSE4261: Neural Network and Deep Learning

Assignment-3

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Question: Discuss feature extraction power of your favorite CNN pretrained by ImageNet dataset before and after transfer learning by MNIST digit dataset by plotting high dimensional feature vectors on 2D plane using three dimension reduction techniques.

Answer:

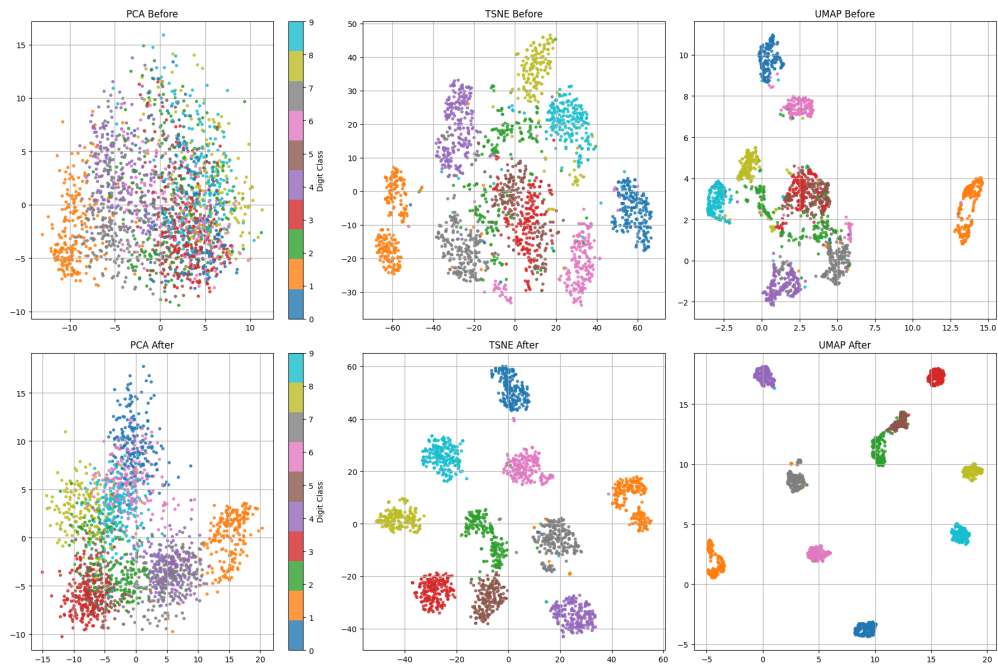


Figure 1: PCA, t-SNE, UMAP of feature vector

Introduction

To perform this experiment I relied on **MobileNetV2** which is a lightweight and efficient convolutional neural network (CNN) trained on the ImageNet dataset. I wanted to test its feature extraction ability after and before the transfer learning on the MNIST digit dataset.

Methodology

I obtained feature vectors of high dimensions (1280 dimensions) out of the MobileNetV2 model. The pretrained version was not fine-tuned initially. Next, I slightly re-trained the network with a smaller dataset (just 5,000 images) of the data-set (MNIST), giving it more specificity to handwritten digit classification.

In order to visualize the extracted feature, I used three dimension reduction techniques:

- **PCA (Principal Component Analysis)** – A linear method that retains maximum variance and shows overall trends in the data.
- **t-SNE (t-distributed Stochastic Neighbor Embedding)** – A non-linear technique that preserves local neighborhoods, ideal for visualizing clusters.
- **UMAP (Uniform Manifold Approximation and Projection)** – Another non-linear method that balances both local and global structure efficiently.

Observation and Discussion

Before transfer learning, MobileNetV2 extracted general-purpose visual features. These showed limited class separation in 2D plots, with visible overlap between different digit classes. After transfer learning on MNIST, the model adapted its internal representations to better capture digit-specific patterns such as loops, curves, and strokes.

PCA revealed more variance explained by class after fine-tuning. t-SNE and UMAP showed significant improvement in class-wise clustering, clearly separating different digits with minimal overlap. This transformation of feature space demonstrates how even a lightweight model like MobileNetV2 can adapt effectively with minimal training.

Conclusion

Using MobileNetV2, I observed a clear improvement in feature representation after transfer learning. The model evolved from general visual understanding to task-specific digit recognition. The dimensionality reduction techniques provided clear 2D visualizations of this improvement. Notably, MobileNetV2 offers this strong performance while being less resource-intensive compared to heavier models like ResNet or Inception, making it ideal for quick and efficient transfer learning on small datasets like MNIST.